

Pushing to Docker Hub

Build once, push once — a 2-step tutorial

This guide shows how to publish (and update) the Gelochip Studio image so anyone can run it with a single ``docker run`` / ``docker compose up`` — no clone, no build on their side.

Docker is NOT Git

In Git you do: `add` → `commit -m "..."` → `push`. Docker has no ``add`` and no commit message. There are only two real steps (plus a one-time login):

```
# 1) BUILD — package everything into an image and tag it
#     '.' = use the Dockerfile in the current folder
docker build -t adeirman2705/gelochip-studio:latest .

# 2) PUSH — upload that image to Docker Hub
docker push adeirman2705/gelochip-studio:latest
```

The BUILD step is the 'add + commit': it bundles your code, deps, datasets and models into one image. There is no message — the TAG (the ':latest' part) is how you label versions.

One-time: log in

```
docker login -u adeirman2705
# paste your Docker Hub password or access token when asked
```

The mental map

Git habits → Docker equivalents

Git	Docker	what it does
git add . + git commit	docker build -t name:tag .	package current state
git push	docker push name:tag	upload to the registry
git tag v1.0	the ':tag' in the name	label a version

Reading the image name

An image name has three parts:

```
adeirman2705 / gelochip-studio : latest
```

```
└─user──┐   └─repo──┐   └─tag┐
```

- USERNAME — your Docker Hub account (adeirman2705).
- REPO — the image name; created automatically on first push.
- TAG — the version label; ':latest' is the default if omitted.

What about docker compose?

`docker compose` RUNS the multi-container stack (app + the Ollama LLM). You do NOT push with it. Because our services use 'image:' (not 'build:'), end users only ever PULL — exactly what you want.

Worked example

Exactly what we run to publish Gelochip Studio

```
# 0) one-time login
docker login -u adeirman2705

# 1) build + tag (run from the repo root, where the Dockerfile is)
docker build -t adeirman2705/gelochip-studio:latest .

# 2) push to Docker Hub
docker push adeirman2705/gelochip-studio:latest
```

That's it. Anyone can now run it:

```
docker run -p 8090:8090 adeirman2705/gelochip-studio:latest

# or the full stack (app + LLM): docker compose up
```

Shipping a versioned release

Tag the same image twice — a frozen version plus 'latest' — and push both:

```
docker build -t adeirman2705/gelochip-studio:v1.0 \
              -t adeirman2705/gelochip-studio:latest .

docker push adeirman2705/gelochip-studio:v1.0
docker push adeirman2705/gelochip-studio:latest
```

Handy commands

```
docker images          # list local images
docker image ls | grep gelo  # find yours
docker rmi adeirman2705/gelochip-studio:v0.9  # delete an old local tag
```

Tip

Re-pushing only uploads the LAYERS that changed — if you edit a little code, the big base/deps layers are reused and the push is quick. Big rebuilds (e.g. swapping CPU→CUDA torch) re-upload more.